

GR-3. 12401067.

Computer Programming using C
Department of CSE
Punjabi University Patiala
MST-2

Time 60 Min

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Section-A (Do All)

- Q.1 Why do we need Arrays? How we initialize a 1-d Array?.....2
Q.2 How we represent end of a string?.....1
Q.3 Write a program in C to show the basic declaration of a pointer.....2.

Section-B (Do any Two) (5 mark each)

- Q.6 How can we pass arguments to a function? Explain in detail.....5
Q.7 Difference between C & C++.....5
Q.8 What do you understand by Recursion? Write a program to display Fibonacci series using

***Note: Do Mention your Groups and Full roll Nos**

MST-II: 11/2023
CSEB1122T: Computer Programming

Time Allowed: 1 Hr

Max. Marks: 15

Note: Attempt all questions in section-A and attempt *any-two* from section-B.

SECTION-A

Q1 (a) How array(s) are represented in memory. Justify your answer with suitable examples. 1

(b) Explaining the following code, predict the output of `printf()` statement.

```
...  
int x=30, y=50;  
justFun(&x,&y);  
...  
void justFun(int* m, int* n)  
{  
    int a=*n, b=*m;  
    printf("a=%d and b=%d",a,b);  
}
```

(c) What values to be printed for A, B and C in the following code segment?

```
{...  
    int X[5]={1,3,2,0,4,5}, *Y;  
    Y=&X[2];  
    printf("A=%d \n", *Y);  
    printf("B=%d \n", *(Y+1));  
    printf("C=%d \n",, *(Y-1));  
}
```

What is the meaning of 'overloading' for any operators and functions. 1

Explain the working of following code segment: 1

```
char* uniName="Punjabi University";  
puts(uniName);
```

SECTION-B

Q2 Giving examples, justify the advantages of sending data references over sending the data directly to a function. 5

Q3 How user-defined structures are different than classes. How data write and read operations are performed in both? Justify your answer by examples of each. 5

Q4 Give example in 'C' to store user-data into files? How data-files are created and read to perform data operations? 5

MST 2

Jan 2022. Common to all branches.

Section A is compulsory. Attempt any two questions from section B. Total 15 marks.

Section A ($5 \times 1 = 5$)

Q1 Define Following Terms

Function, Array, String, NULL Character, Class

Section B Do Any 2 ($5 \times 2 = 10$)

Q2) Explain important concepts of object oriented programming.

Q3) Write a program to calculate factorial of a no. using function.

Q4) Write a simple program which shows the necessity of structures.

Q5) Explain various File opening Modes and write a program to copy one file into another.